Database Design for Flight Attendant Training & Certification

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Project Context

- Flight attendants must complete regular safety trainings
- Certificates expire and must be renewed
- Everything has to be documented properly

Drawing from my previous experience in aviation, I wanted to model a real-world process where safety, compliance, and documentation play a central role.

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Project Objective

- Design a clear ER model
- Represent real training processes
- Avoid duplicate or inconsistent data
- Convert the model into working database tables

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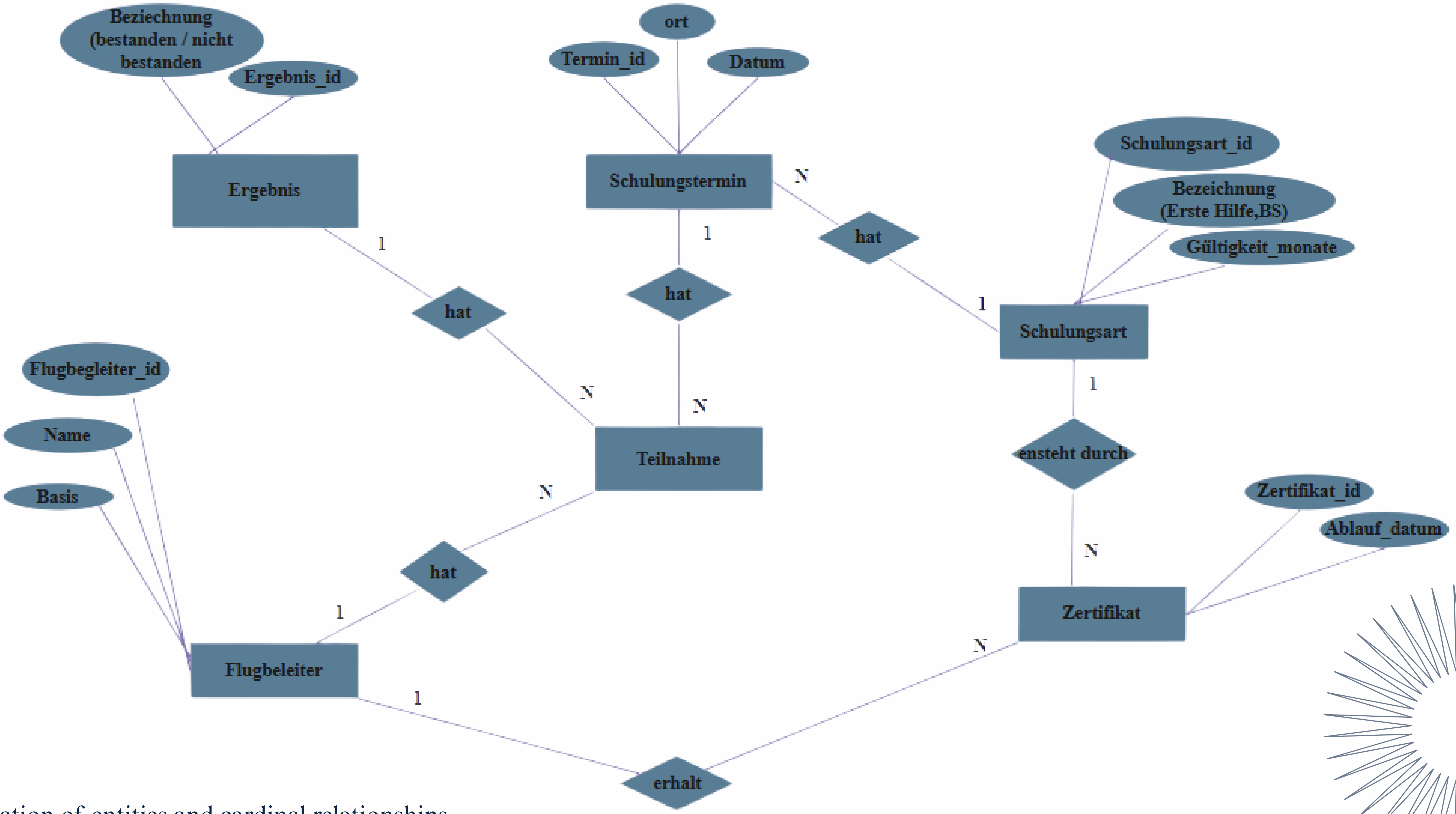
ER-Model and Relationships



Conceptual representation of entities and cardinal relationships.

Resolving the M:N Relationship

The Problem: Many-to-Many Relationship

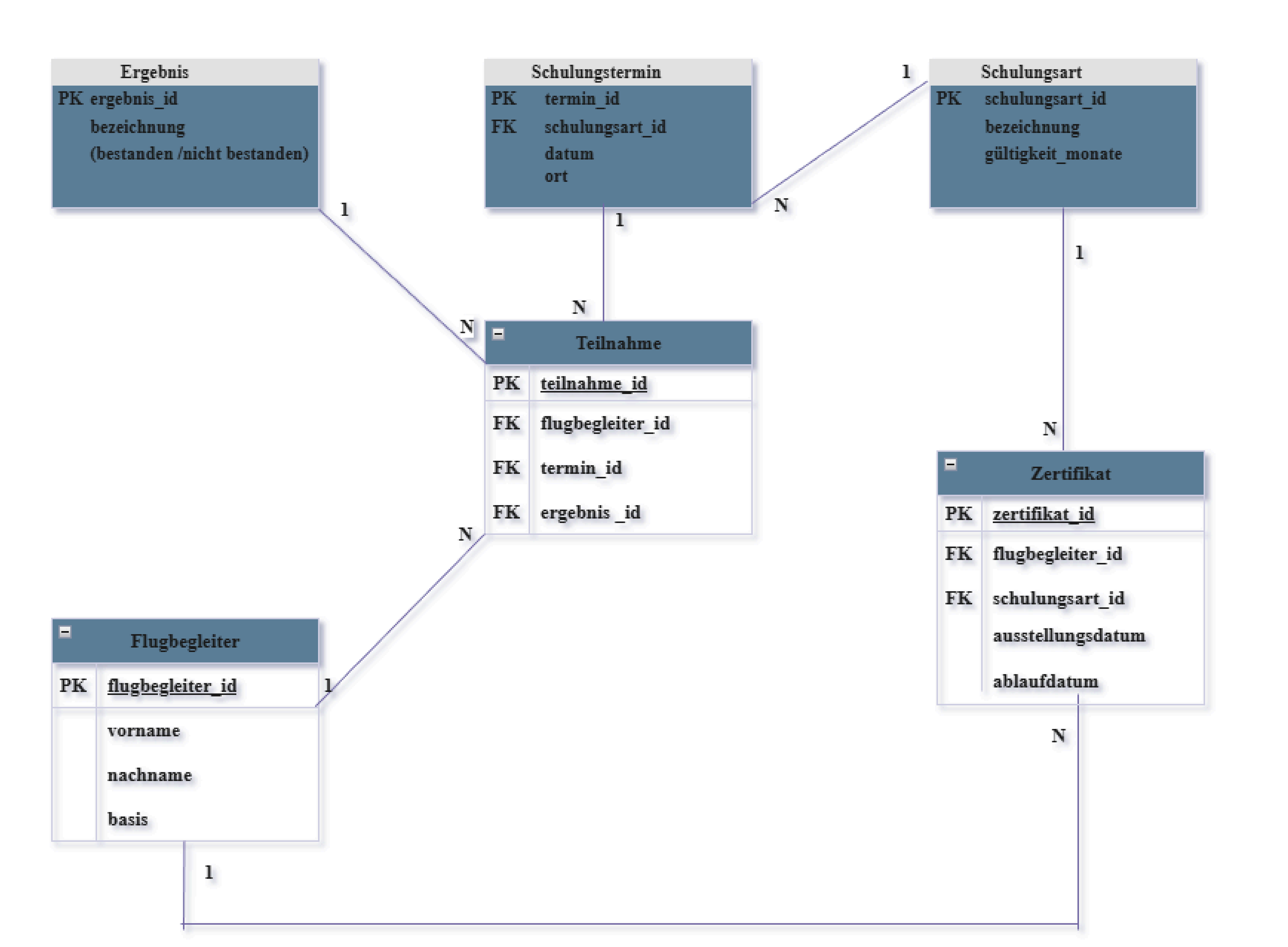
- A flight attendant can attend several training sessions
- A training session can include several flight attendants
- This creates a many-to-many (M:N) relationship
- In relational databases, this cannot be implemented directly

The Solution: Participation Table

- I introduced a junction table called "Participation"
- Each record connects one flight attendant with one training session
- The training result is also stored in this table
- The M:N relationship is transformed into two 1:N relationships

Relational Database Implementation

- Each entity becomes a table.
- Primary keys uniquely identify records
- Foreign keys connect the tables
- This keeps the data consistent and structured



What I Learned & Why It Matters

- Clean modeling prevents data redundancy
- Proper key design ensures consistency
- Real processes can be translated into structured database logic

This project helped me connect my aviation discipline, safety standards and structured procedure with database design principles.

Thank You for Your Time

Questions and feedback are welcome.